

1 Methods

To simulate membranes, we used a triangular mesh. Each timestep, the vertex positions are updated based on the discrete membrane Hamiltonian using different integration schemes. Our simulator calculates the gradients and Hessians of the Hamiltonian based on the geometric information around the vertex. To model the membrane we use the Helfrich Hamiltonian, its most general expression is.

$$H_{\text{Helfrich}} = \int_{\mathcal{M}} \left(\frac{\kappa}{2} (2H - c_0)^2 + \kappa_G K + \sigma \right) dA \quad (1)$$

Where H and K correspond to the scalar mean curvature and Gaussian curvature respectively. The constant c_0 corresponds to the spontaneous curvature. The constants κ, κ_G and σ are the bending rigidity, the Gaussian modulus and surface tension. For the simulations we need to calculate the discrete gradients of the energy. Therefore we need to discretize the integrals. Our discretization scheme consist of using the dual area of each vertex and using local information to calculate the approximated quantity. For the squared mean curvature term, this looks like

$$\frac{\kappa}{2} \int_{\mathcal{M}} (2H - c_0)^2 \approx \sum_{v_i \in V} (2\hat{H} - c_0)^2 A_{v_i} \quad (2)$$

At this point one needs to make the decision for the discrete scalar mean curvature $\hat{H}$ and the dual area A_{v_i} expression. Based on [reference] one reliable discretization for the mean curvature is.

$$\hat{H}_i = \sum_{e_{ij} \in N(v_i)} \frac{\ell_{ij} \varphi_{ij}}{4} \quad (3)$$

where e_{ij} is the edge that connects vertex i and j and $N(v_i)$ is the neighborhood of vertex i , ℓ_{ij} is the edge length of edge ij and φ_{ij} is the dihedral angle between the two faces that touch the edge ij . Its important to notice that since our mesh is manifold, every edge that is not in the boundary has only two faces touching it.

2 Visualization

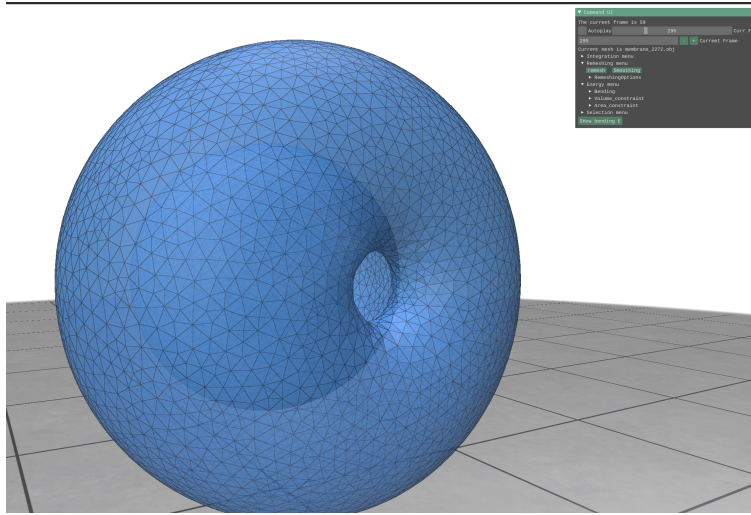


Figure 1: Polyscope interface